Best Route VS. Best Emissions

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Agenda

Introduction

What: an AI system for any city trip that compares the fastest route vs. the lowest emissions route which advises a speed to catch only green lights.

Problem: navigation optimizes for time, but stop and go idling causes most urban emissions. Is the fastest route also the cleanest?

How: forecast traffic from historical volume and signal timing, then route to minimize stops and emissions.

Related work: graph based traffic forecasting, eco-routing, and Green Light Optimal Speed Advisory (GLOSA).

What is our Motivation? Why do emissions matter?

Why this project?

Congestion and idling causes a large share of urban emissions. Routing is a cheap and infrastructure free lever.

Impact & Benefits

Lower CO_2 emissions, better fuel economy and EV range, more reliable travel times, and a measured time vs emissions trade off.

Who does this benefit?

Drivers; autonomous delivery/rideshare fleets; cities and public health; traffic engineers.

Why should people care about emissions?

Roadside pollution near intersections harms health, and small routing/speed changes compound across millions of trips.

Dataset

Link 1: LargeST — github.com/liuxu77/LargeST

Link 2: BasicTS toolkit — github.com/GestaltCogTeam/BasicTS | SUMO — eclipse.dev/sumo

Dataset 1 — Traffic (LargeST Bay Area Only)

Thousands of sensors, 2017–2021 speed/volume, used to train the traffic forecaster.

Dataset 2 — Signals and Emissions (SUMO)

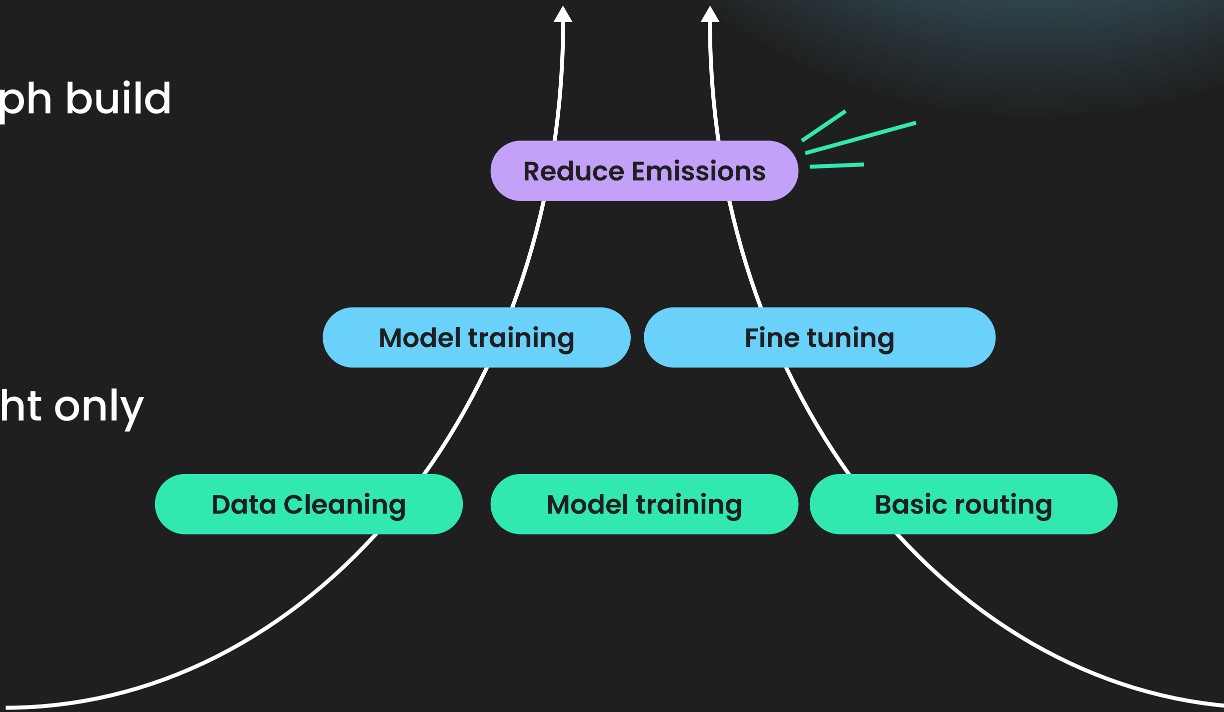
Supplies signal timing, per-car emissions and a testbed to measure routes with GLOSA.

Why two datasets?

No public dataset has traffic signals and emissions together. Dataset 1 gives real traffic, Dataset 2 supplies signal timing and emission labels.

Model Plan

1. Data cleaning & road-graph build
2. Traffic forecasting
3. Baseline routing (fastest)
4. Emissions model
5. Best emissions + green light only routing
6. Validation + Results



Model Plan

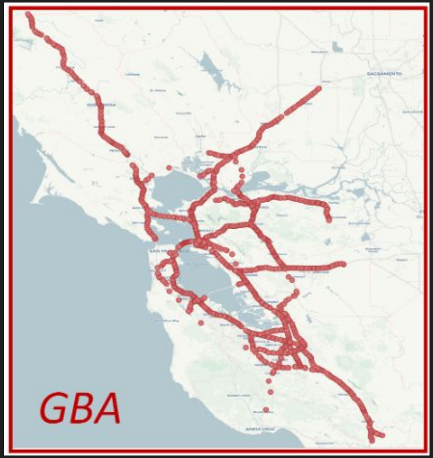
Framework: BasicTS - a toolkit that LargeST is built on (runs on PyTorch).

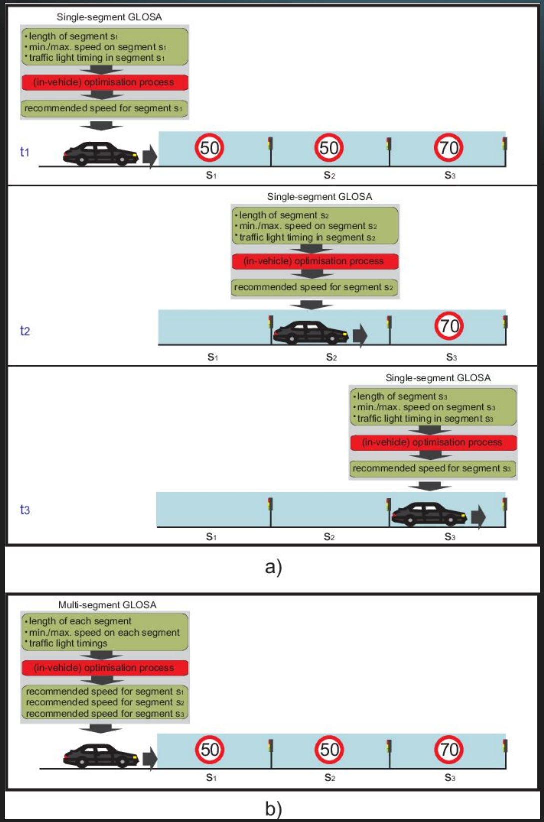
Forecasting: STAEformer (Spatio-Temporal Adaptive Embedding transformer) provided by BasicTS.

Routing: A* search on an OpenStreetMap graph to measure weighted travel-time vs. emissions cost.

Emissions: Read directly from SUMO's built-in emission model. STAEformer predicts traffic speed while SUMO simulates cars emissions

Speed advisory: Green Light Optimal Speed Advisory (GLOSA) a deterministic Python/SciPy calculation.





Model Plan

Routing

Shortest distance path
Fastest route on historical average travel times

Key test: does reduced emissions routing beat normal navigation?

Forecasting

Historical Average for baseline
Using built-in STAEformer model from BasicTS framework for predicting traffic flow

Emissions

Planned route estimate vs. SUMO emissions calculation

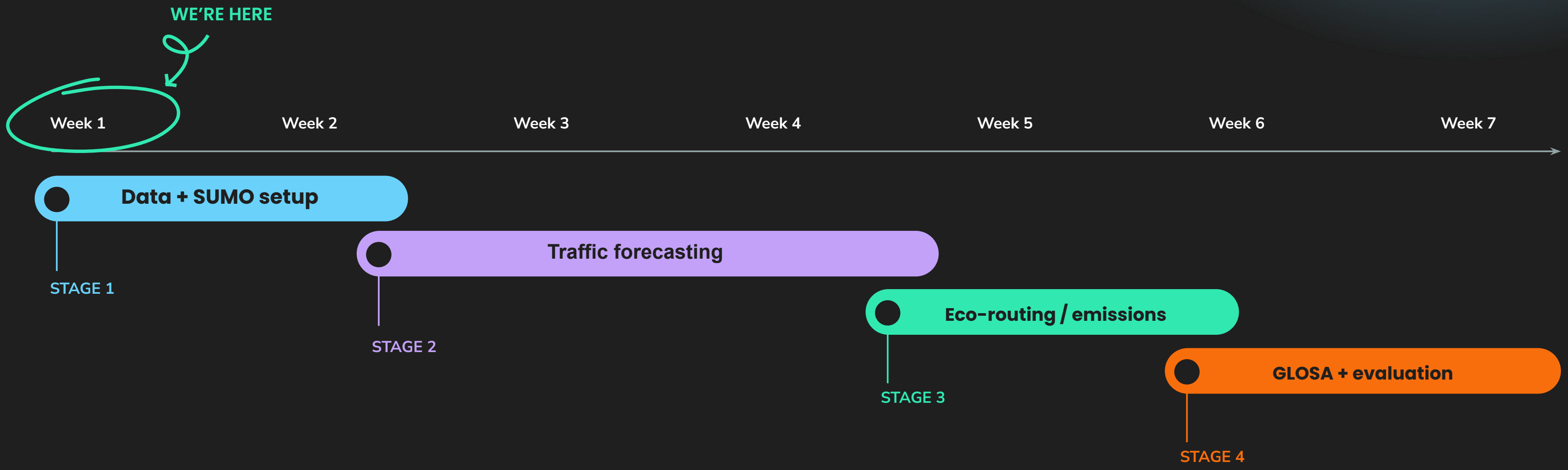
Shows why modeling stop and go matters

Validation

Drive the suggested route / no GLOSA vs. GLOSA green light speed

Compare stops, idle time, and emissions

Timeline


WE'RE HERE
Week 1
Week 2
Week 3
Week 4
Week 5
Week 6
Week 7
Data + SUMO setup
STAGE 1
Traffic forecasting
STAGE 2
Eco-routing / emissions
STAGE 3
GLOSA + evaluation
STAGE 4

Risk Analysis

Risk 1

Signal timing and emissions are missing from public data, have to tie both datasets together.
Mitigation: OpenStreetMap

Risk 2

Simulations don’t necessarily mimic reality in all cases, not a one size fits all solution. Limited to only the Bay Area.
Mitigation: Swap dataset and retrain the model.

Risk 3

At a critical mass no amount of optimization can improve routes.
Mitigation: The routing engine must include a dynamic fallback threshold.

Thank you!